Just Dance Custom Music

!(Translated using a translator)!

Just Dance Custom Music is an independent fan project aimed at studying and creating custom music content for the Just Dance game series.

The project offers a practical algorithm for selecting suitable music, preparing audio files, and creating synchronized karaoke content using custom timing data.

It is intended for enthusiasts who want to learn how to prepare custom music content and integrate it into the existing game structure.

Author **sf-brat**. URL project - https://github.com/sf-brat/Just-Dance-Custom-Music-/

!Attention! This project is an unofficial fan-made development. It is not affiliated with Ubisoft and has not been approved by the company. Just Dance and the corresponding game materials belong to their respective rights holders. The musical works belong to the respective authors and rights holders.

Instructions for replacing the original song with a custom one

What you will need:

0) The “Just Dance” game, which has access to system files.

1) music_selection.html – an HTML file for selecting music.

2) karaoke_create.py – a Python file for creating and editing karaoke for the game.

3) A unpacker and packer for .ipk files (I use “ubiart-archive-tools” from the user PartyService (https://github.com/PartyService/ubiart-archive-tools).

4) Python installed on the PC and the required dependencies (requirements.txt) installed.

5) The original clip we’re interested in, where we want to replace the song.

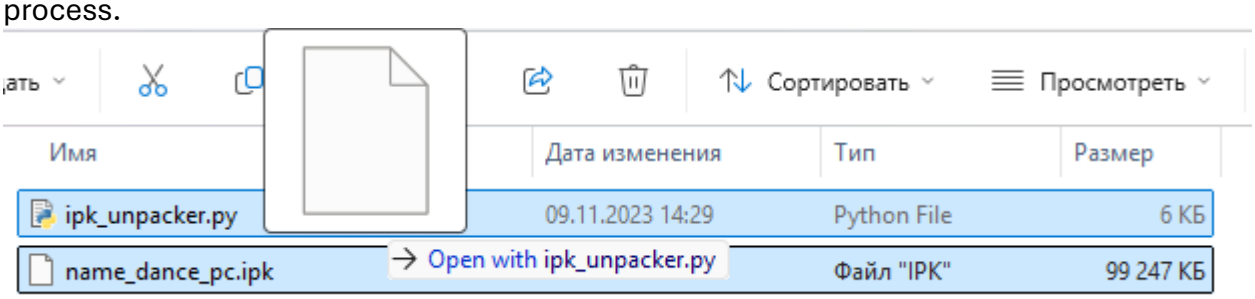
6) A list of songs we’d like to add (candidates).

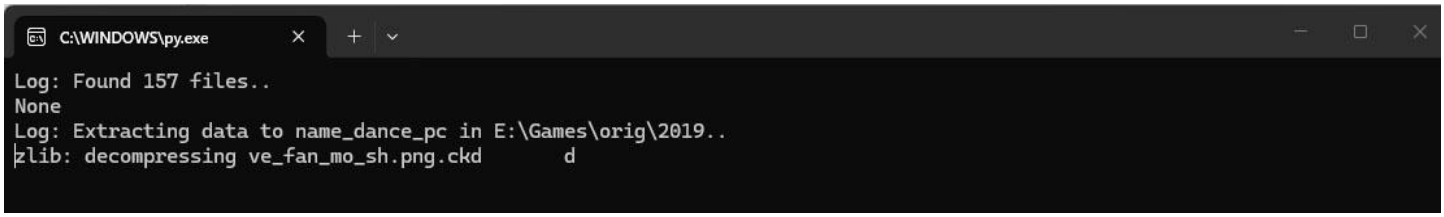
7) Any program for working with video and audio.

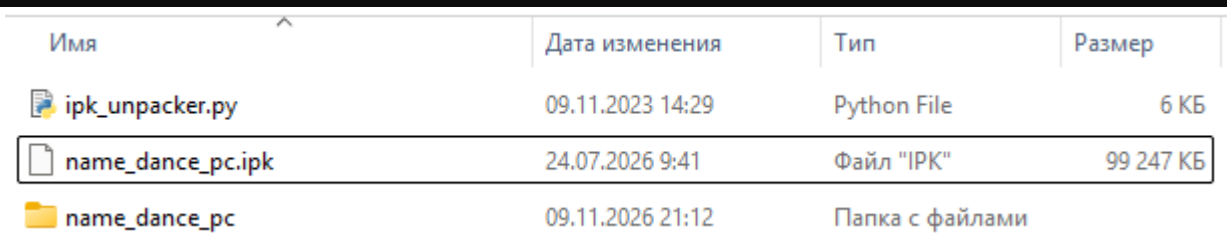
Action plan:

1) Find the clip you’re interested in, where you want to replace the song.

2) Unpack the archive using the “ubiart archive tools” utility (PartyService) – drag the file onto the ipk_unpacker.py file. A folder with the same name will be created in the location where the archive is located (dance_pc.ipk -> dance_pc (folder)). The “Command Prompt” will open, showing the archive unpacking process.



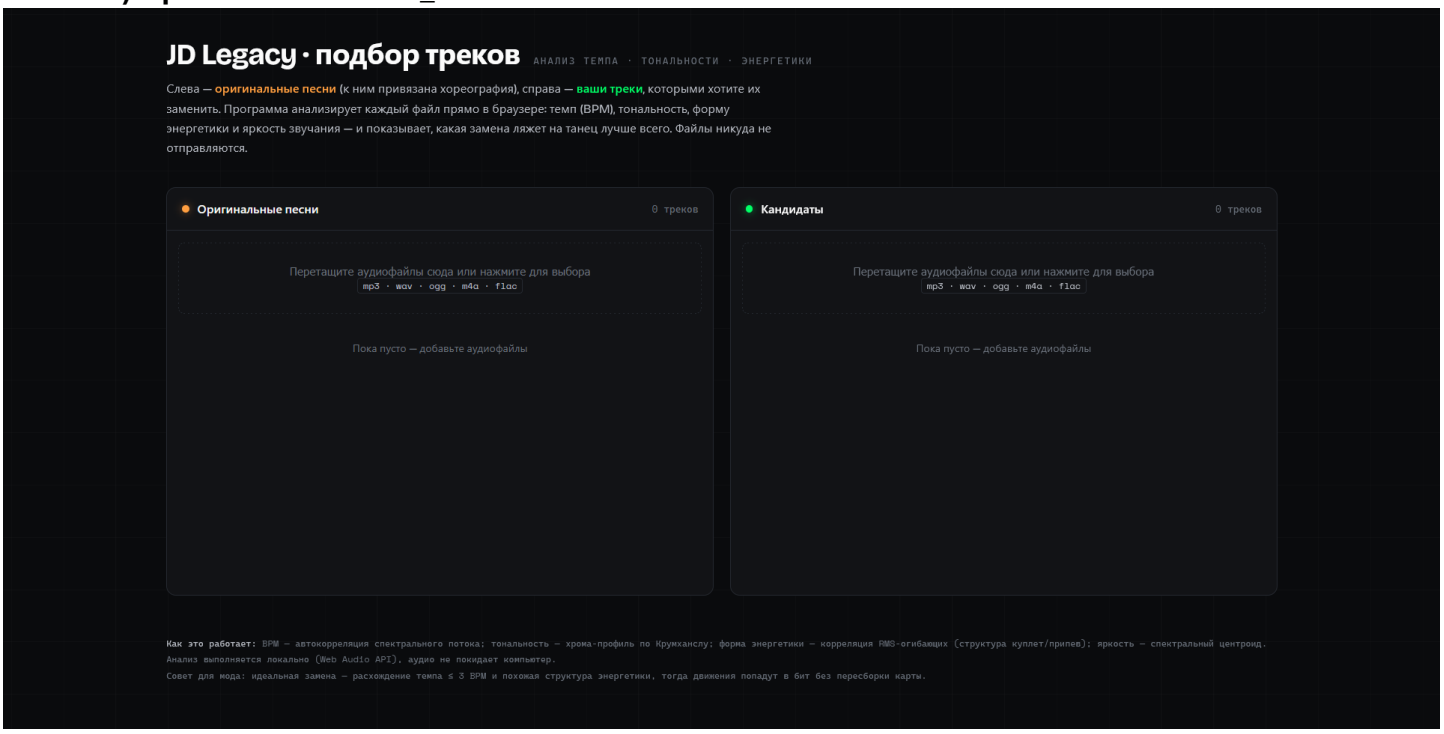




3) Analysis of the original song with candidates

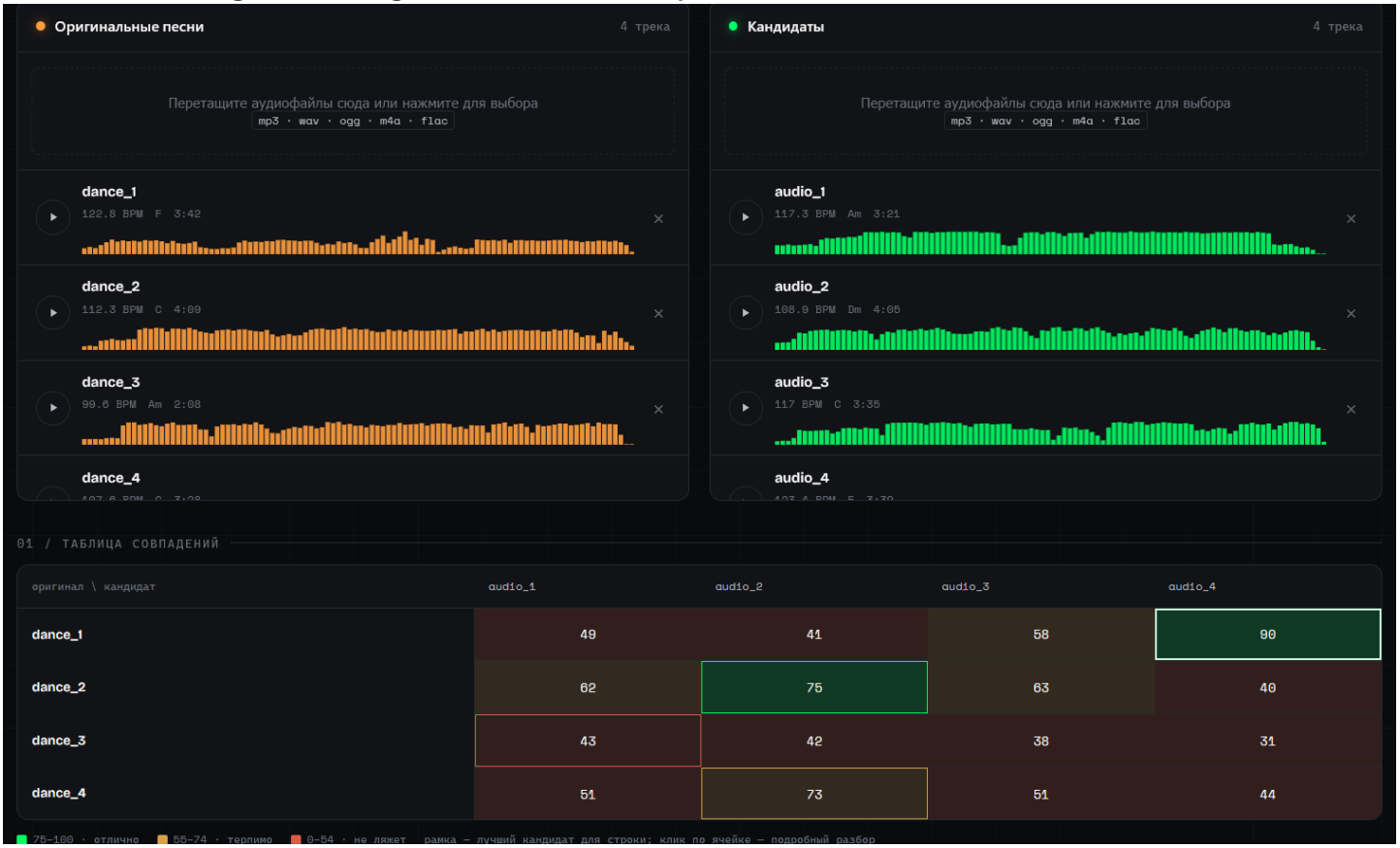
3.1) Go to the path – ...name_dance.../world/maps/...name_dance.../audio

3.2) Open the file music_selection.html



3.3) Load the original songs from the unpacked file into the left column, and your own songs into the right column. After analyzing all the songs, the website program will display a table with the percentage of matches — the higher the percentage, the higher the chance of a replacement. And if a cell has a border, it means it's an "Ideal Replacement" — the tempo discrepancy is ≤ 3 BPM and the energy structure is similar; the movements will fit into the beat without reassembling the map.

After loading all the songs of interest, we analyze the table.



75–100 отлично (excellent) | 55–74 терпимо (tolerable) | 0–54 не ляжет (unacceptable)

3.4) If we click on any cell, a block with a detailed comparison of the songs “Original” and “Candidate” will appear below. First of all, we look at the parameters “Tempo, BPM” (the most critical) and “Duration” (also the most critical). The parameters of medium importance are “Key” (to ensure the song has a single key and to increase the likelihood of beat matches).

We also take the “Energy” chart into account. Here, we can immediately identify the “bursts” and “downtimes” of the song.

The conclusions from the dry analysis are also shown below all the graphs.



There is no 100% exact match here, as the dance was created specifically for the “Original Song”. And the chance is small that both the voice, the beat, and the energy will be 100% aligned. But overall, the general comparison of the songs works quite well.

3.5) We select the more suitable songs based on the table. For example, based on the illustration, for the “dance_1” dance, “audio_4” might be the best fit, and for “dance_2” -> “audio_2”.

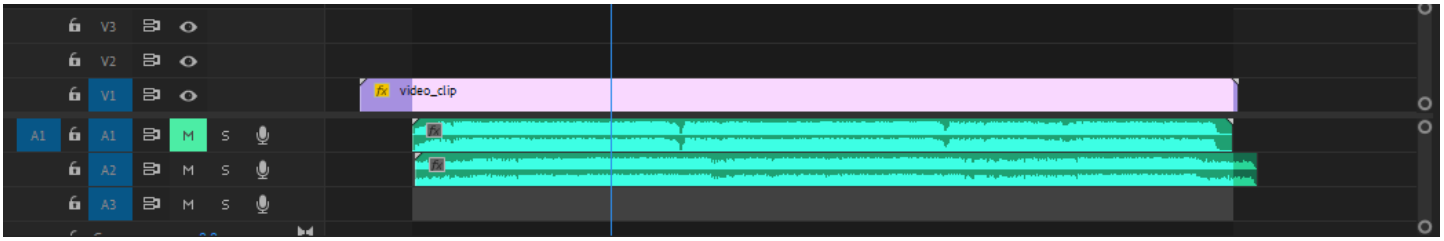
4) Matching the candidate song to the original clip

4.1) Open the video editor and load the video clip from the “videoscoach” folder and the audio from the “audio” folder into it.

4.2) Match the audio with the video clip so that the start and end of the dance are exactly aligned. We rely on the “silence” in the audio or on the end of the dance.



4.3) After comparing the original files, we now load the candidate for this clip. Visually align the original and candidate by bitrate. After that, we can disable the channel with the original audio track.

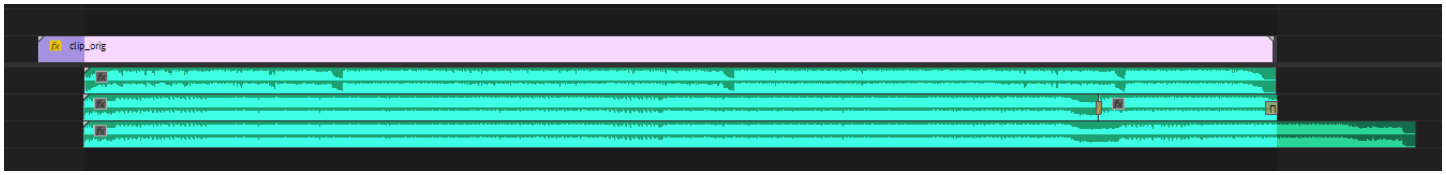


Let’s take clip_original – audio_custom as an example.

4.4) We watch and listen to the clip in full. We check that the beat matches the dance, that the dance’s energy is in sync, and that the clip looks good/great visually, and that there is no “desynchronization” between the dance and the song (the dance has a “lull,” while the song continues with speech or features a “heavy beat”).

Even if the verse/chorus of the original song differs from our custom song, it might still fit perfectly with the dance, and this “mismatch” won’t be noticeable.

If the custom song is 10–20 seconds longer than the original, you can shorten the beginning and the lull, but make sure it doesn’t cut off on the words. But also be prepared that you’ll have to adjust the beats and mix the song properly.

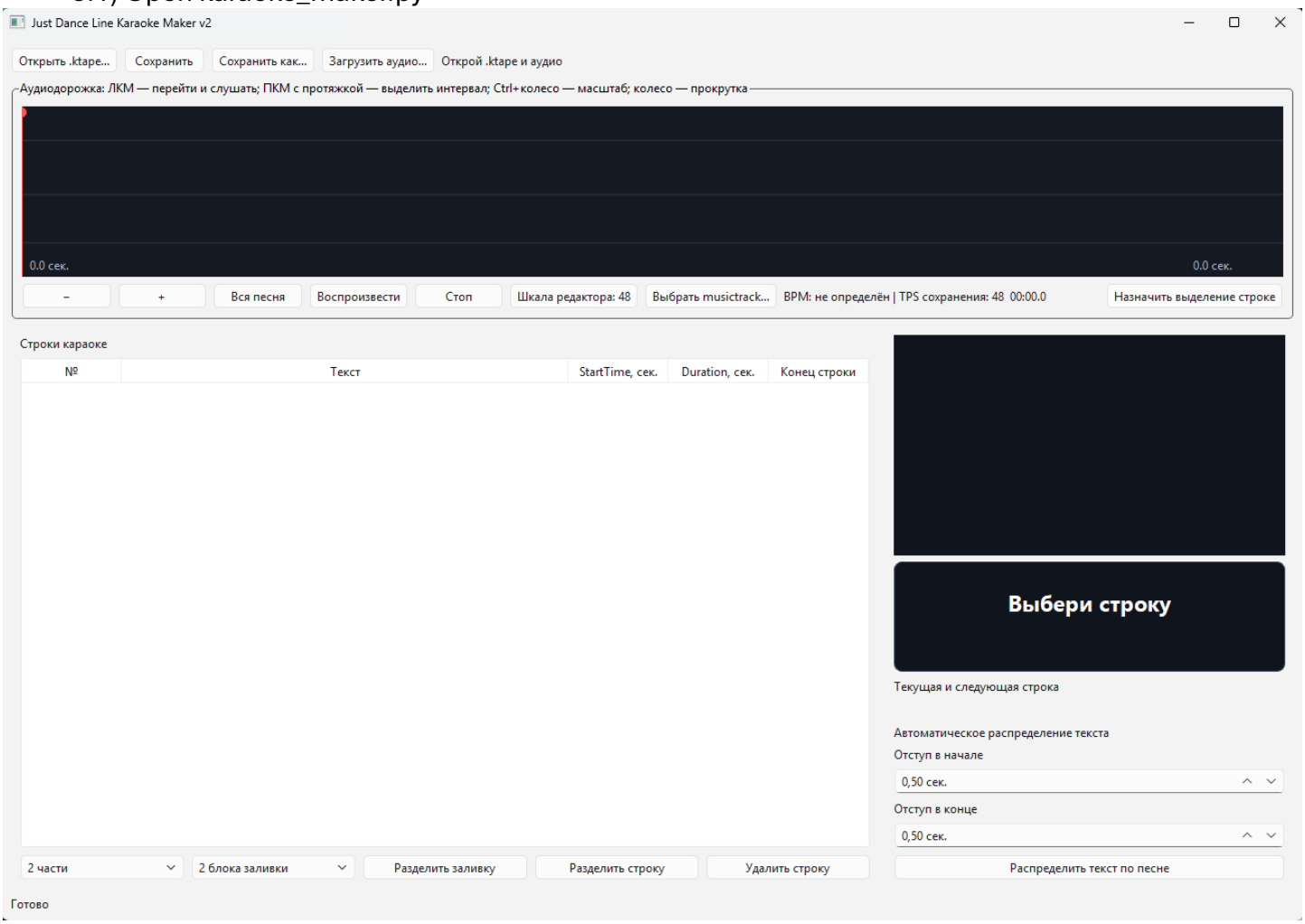


In the photo above:
1 audio line – the original song,
2 audio line – a custom song adjusted to match the dance,
3 audio line – a custom song without any adjustments (full length).

4.5) If there is an exact match in terms of beat and no dissonance is observed when viewing, we have found an exact replacement. We save the audio clip with exactly the same length as the original (and not a millisecond less) in .ogg format. We name it the same as the original. We replace the file. (...name_dance.../world/maps/...name_dance.../audio/...name_dance...ogg)

5) Adding your own karaoke to the modified clip

5.1) Open karaoke_maker.py



5.2) Open the .ktape, audio, and .musictrack files.

«Открыть .ktape» (“Open .ktape”) – …name_dance…/cache/itf_cooked/pc/world/maps/…name_dance…/timeline/ and select the file “…name_dance…_tml_karaoke.ktape.ckd” (we can skip loading it and start from scratch).

«Загрузить аудио» (“Upload audio”) – a custom song that has already been created and saved.

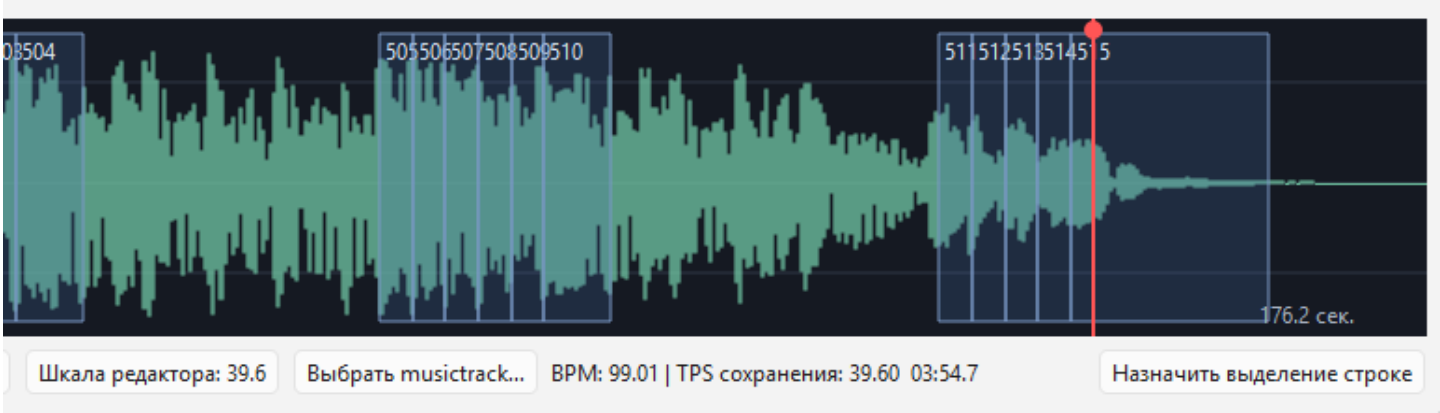
«Выбрать .musictrack» (“Select .musictrack”) - …name_dance…/cache/itf_cooked/pc/world/maps/…name_dance…/audio and select “…name_dance…_musictrack.tpl.ckd”

5.3) After selecting all three files, the screen will display karaoke (ktape), audio, and TPS parameters.



How to work with TPS?

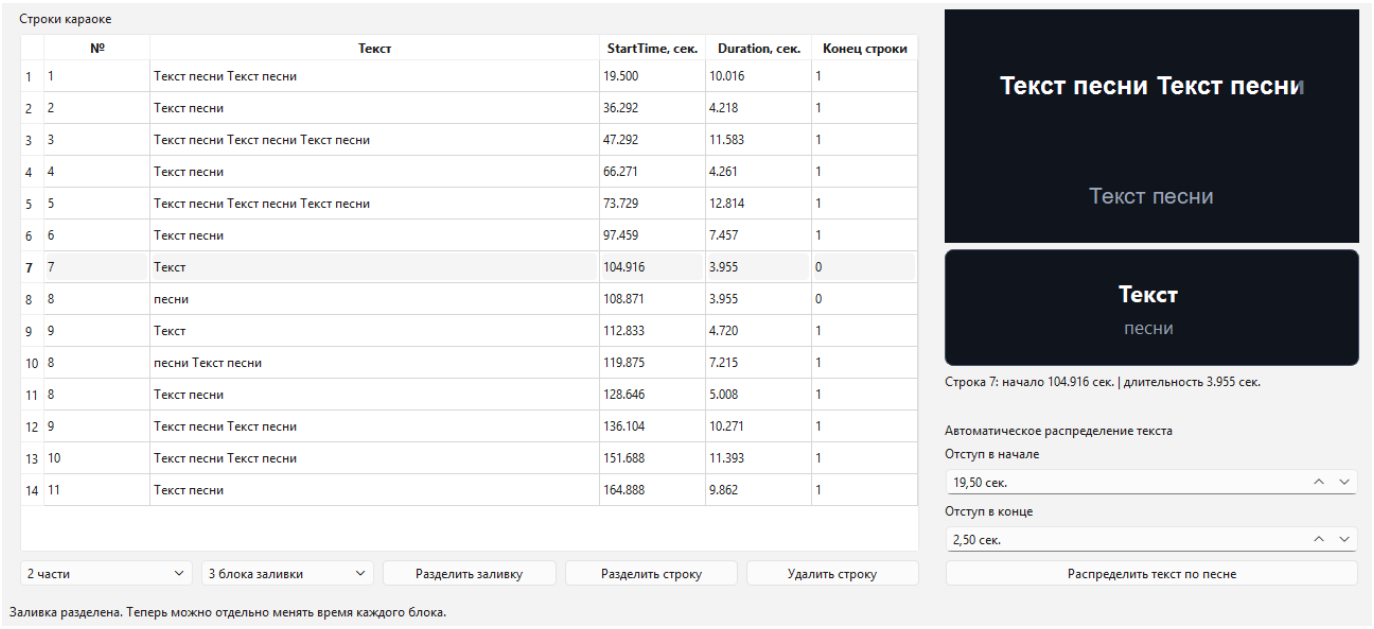
To work with your own text, it is recommended to use the TPS value of 48, since when saving, the program takes into account the TPS value of the original song and automatically adjusts to their value. We change it to check whether the program has identified it correctly and to compare whether the karaoke text matches the original audio.



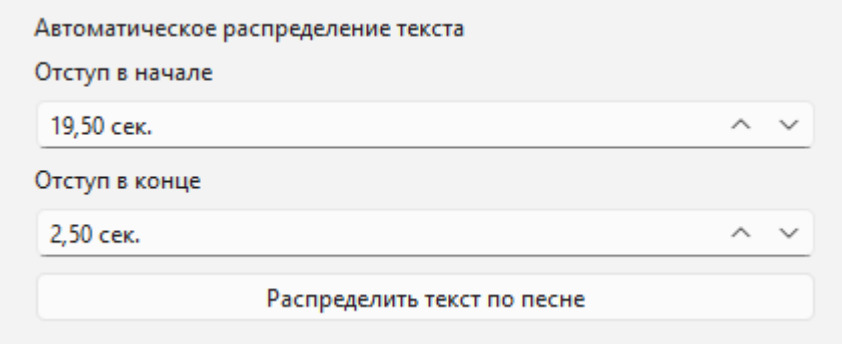
In the photo, we set the TPS to the value determined by the program and compared the last words with the song. If they match, that’s great; we return the TPS back to the value of 48.

5.4) How to create your own karaoke text?

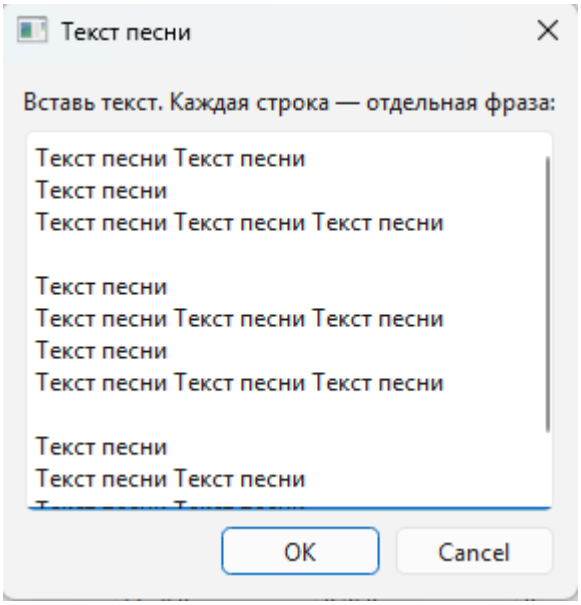
After you’ve uploaded all 3 files, the karaoke text of the original clip will appear at the bottom of the program. Here, when you move the pointer, an approximate karaoke playback is played (the right part of the program) and the text breakdown itself is displayed (the left part of the program).



In the bottom right corner of the program, there is a window called “Automatic text distribution” (**«Автоматическое распределение текста»**). With its help, we will be able to add the text from the song. We determine at which second the voice starts in the custom song and how many seconds to leave for the last word (but this is not mandatory). And then we click “Distribute text across the song” (**«Автоматическое распределение текста»**).

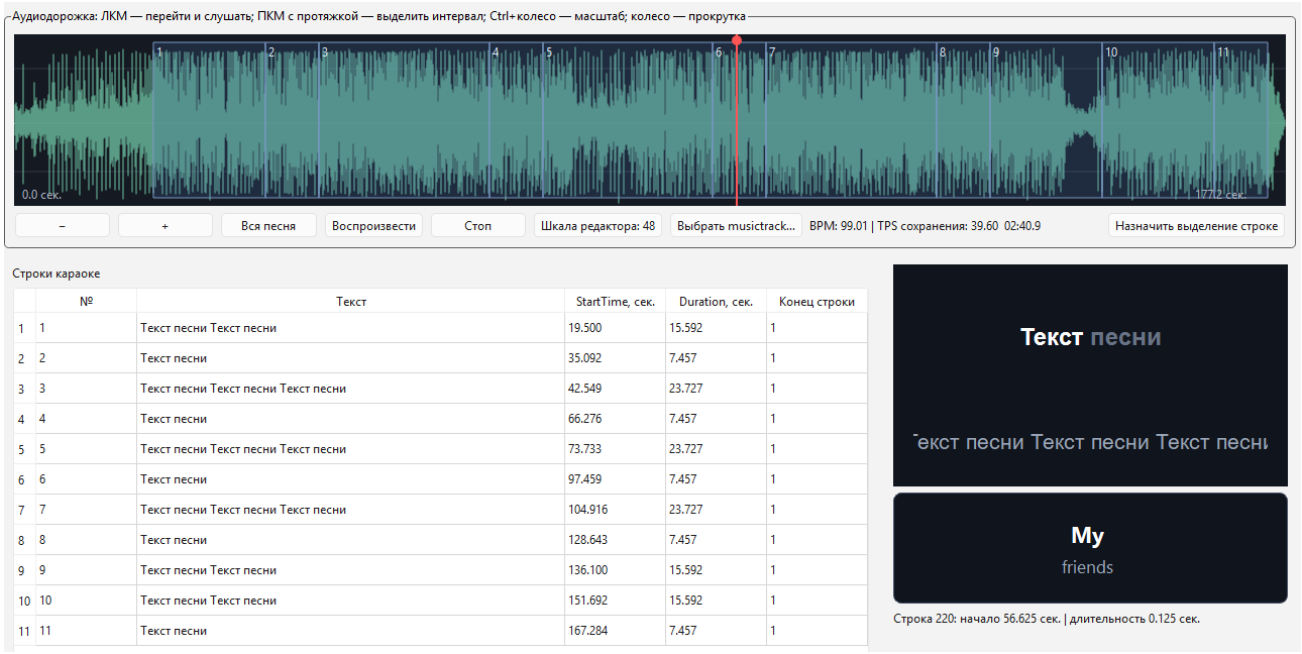


A window opens with an input field for the song lyrics. On the Internet, we enter “ “Author and song title” song lyrics”. We find the correct lyrics for the song and copy them. We paste them into the window in the app and click “OK”.



After that, the program distributes all the lines throughout the entire musical composition!

You will need to manually edit each block.



What do the columns mean?

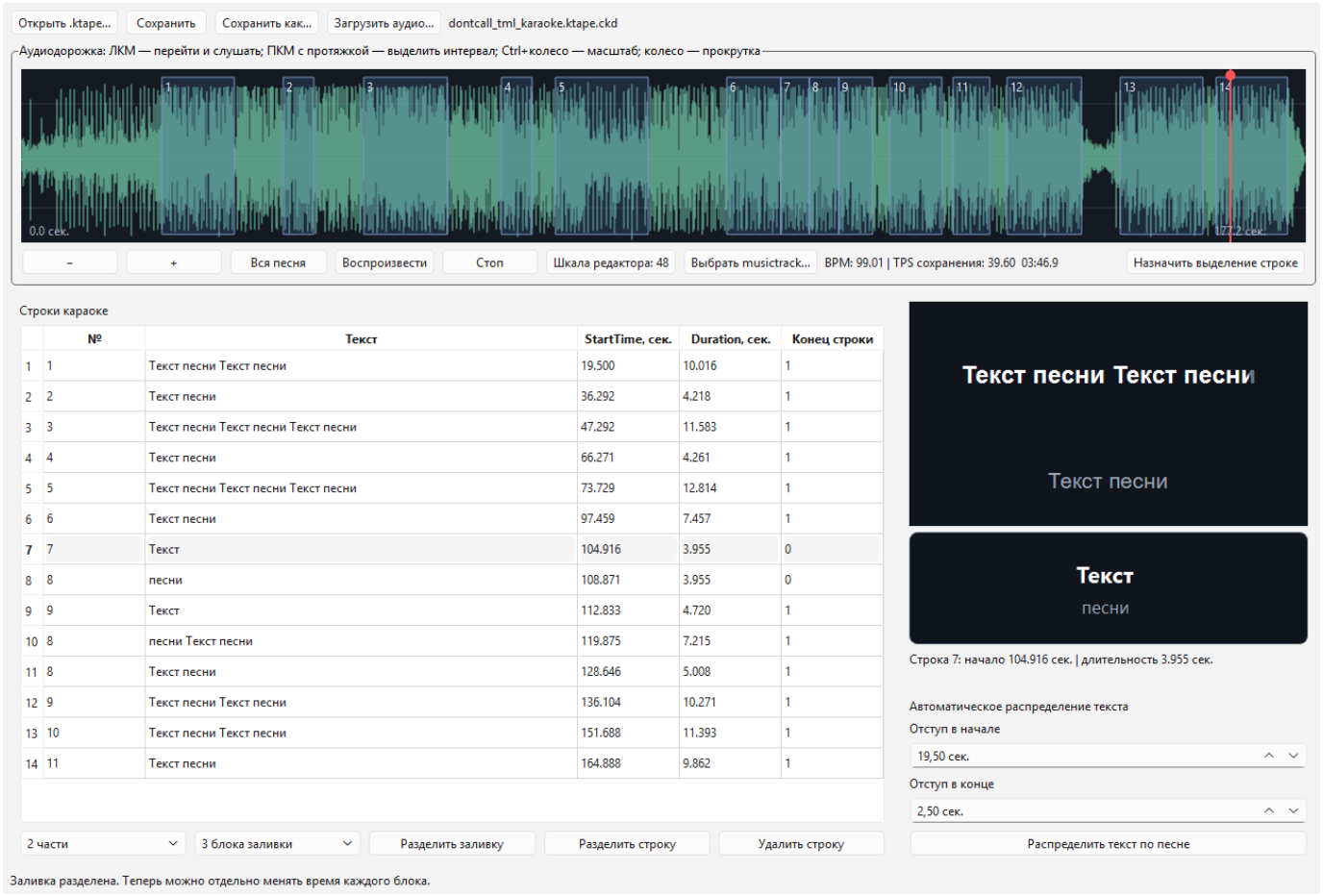
№ – line number (this is not editable; the program numbers them automatically)

Текст (Text) – the karaoke text itself. It can be either a single letter/symbol or a whole sentence.

StartTime, sec. – the start time of playback for this line.

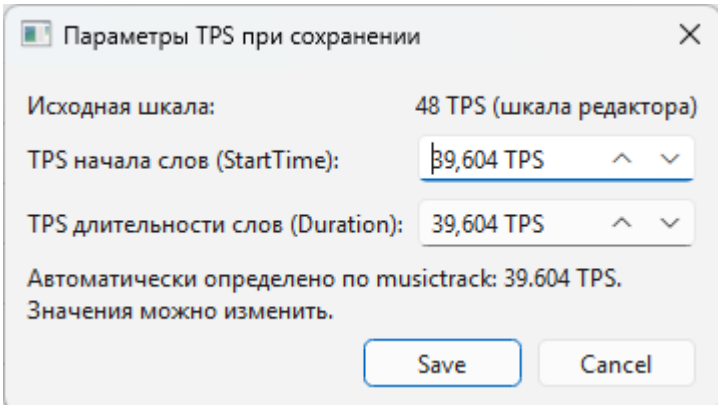
Duration, sec. – the duration of playback for this line.

Конец строки (**End of line)** – indicates that this text is the last one in the line (1 – end of line, 0 – continuation).



On the graph, the music can be dragged left/right beyond the text frame for better adjustment. If the custom song has parts where the beginning of a sentence is spoken quickly, and the end is slower, you can split the line using the **“Split Fill” («Разделить заливку»)** button.

5.5) When the entire song text has been correctly aligned with the visual elements and all the words match when listening, it's time to save the karaoke. Click "Save" and "Save". The file itself is **"...name_dance..._tml_karaoke.ktape.ckd"** is being re-saved; no manual action is required.



After creating your own karaoke, we move on to the penultimate stage.

6) Editing the video title

6.1) Go to the path ...name_dance.../cache/itf_cooked/pc/world/maps/...name_dance.../ and open the file "songdesc.tpl.ckd" using any text editor (Notepad, etc.)

6.2) From the long string, we are interested in two values: "Artist" (Song Author) and "Title" (Song Title) Here, we can replace the values with our own ones ("Artist": "author_original" -> "Artist": "author_custom"), and also with the song title.

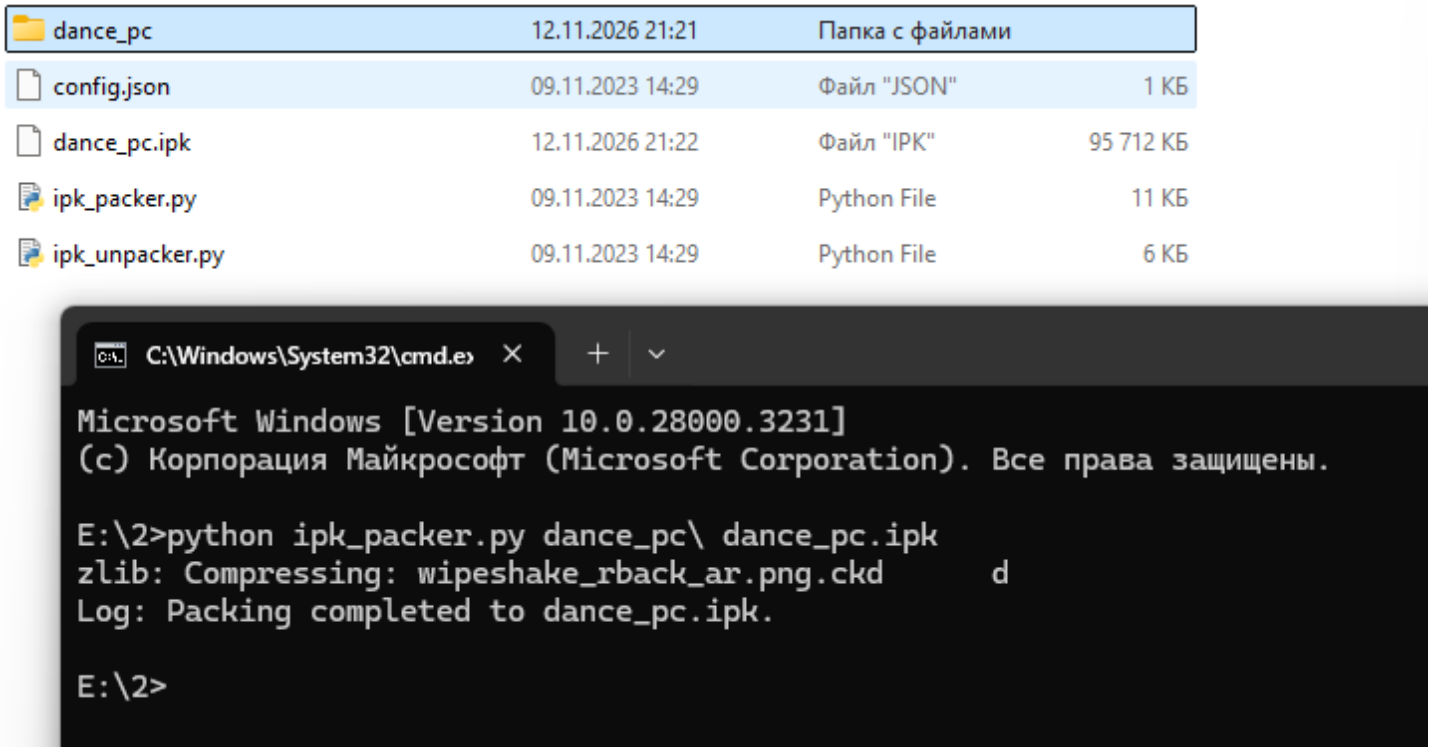
```
,"Artist":"Custom_author","DancerName":"Unknown Dancer","Title":"Cutstom_title",
```

6.3) Save the changes. Now we can move on to the final stage – archiving and checking in the game.

7) Archiving and checking in the game

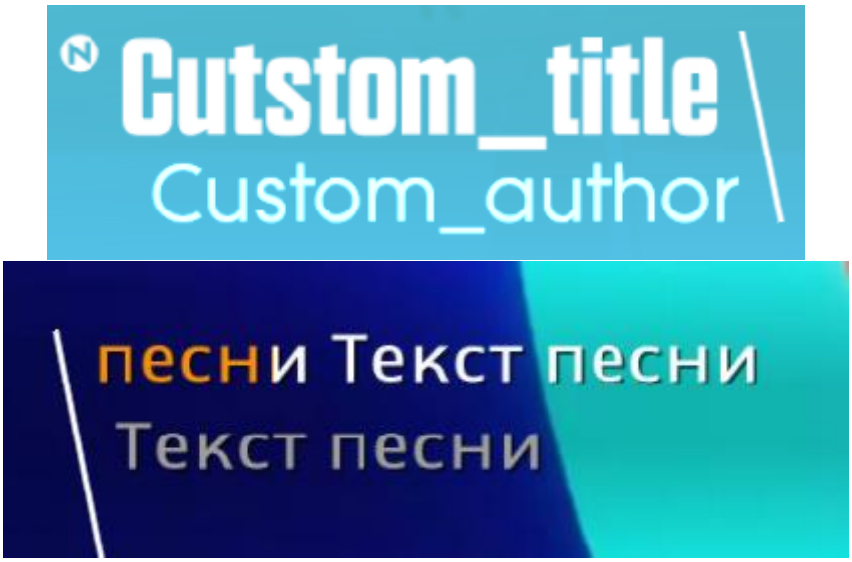
7.1) After all the changes, go back to the root of the folder containing the ipk_packer.py file and the "...name_dance..." folder. Open the console in this location.

7.2) Enter "python ipk_packer.py dance_pc\ dance_pc.ipk" (replace "dance" with the name of the clip).



7.3) After saving in .ipk, move it to the same location where you took the original clip (save the original clip so you can restore it to its original state).

7.4) Launch the game and find your modified clip. It will play a custom song, the name will be changed (if it was changed), and when you play the dance, karaoke will be displayed.



If the music in the game itself is playing great and the karaoke is going well, great!
We did everything right.
And so you can replace any clip.

Examples (YouTube)
- https://youtu.be/FVFZLaV1Z58
- https://youtu.be/0Z799sijip0
- https://youtu.be/xNbnkS6snss
- https://youtu.be/x-vachbyn_Q

YouTube a channel with my examples of replaced songs -
https://www.youtube.com/channel/UCxfQ2b-UlgkmAm45u5P4Tcw/

Mistakes

*** The circle is spinning and the song is not ending –** I have encountered this problem myself, but I do not know the solution.

***The karaoke text does not match the song –** you need to work in the editor with tps 48, but save it with the value of the original song **(step 5.5).** But I came across one clip (maybe there are some more), in which the first half matches, and the second half goes awry. A decision will be made later.

(Perhaps the list will be expanded)